RTAB-Map Back End: Simulation and Real-World Configuration

Engineering note for Chapter 3 (implemented system) and Experiment 4 (dense map)

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Reporting cutoff: 2 November 2026

Purpose of this document. RTAB-Map [1] is the mapping back end of Pipeline 2. It runs behind a front end that supplies the pose, and it is configured differently in the two domains because the two camera rigs differ. This note records both configurations as they are implemented at the cutoff, the measurements each setting rests on, and the validity checks that decide whether a real-rig map can be trusted. Every number traces to a file named in section 4. Experiment 4 results are not restated here beyond the figures needed to justify a setting.

Status: completed. Simulation configuration. Construct, extend and localize modes are implemented and exercised; the tiny-map and realistic-world databases were built and scored against ground truth (Experiment 4, simulation slot). The configuration has been stable since 9 September 2026 apart from the `loop_closure` switch added on 26 October.

Status: partial. Real-world configuration. The fisheye rig is rectified in the shim and two recordings (28 and 31 October) produced databases, so the chain runs end to end. Two items are open: the 31 October recording needs a chequerboard calibration of the moved mount (a self-calibration from the bag stands in); and no independent ground truth exists for the real site, so no map-quality metric can be reported for either database. The IMU height above the floor was measured on 4 November (0.11 m).

1 What is common to both domains

RTAB-Map never receives odometry as a topic. It takes the pose from TF, so the front end is whichever node publishes the `odom_frame` to `body_frame` transform. Swapping ORB-SLAM3 [2] for VINS-Fusion [3] changes nothing in the back end. The measurement behind this choice is in table R.2: with odometry on a topic the five-way synchroniser evicted the pose before its stereo pair arrived and the run kept three nodes; from TF the same bag gave 28.

Both domains run the same five packages of the wrapper fork (`rtabmap_msgs`, `rtabmap_conversions`, `rtabmap_util`, `rtabmap_sync`, `rtabmap_slam`) against the apt core library `ros-humble-rtabmap` 0.23.7. The fork is pinned to upstream `2eef2b3`, the newest ROS 2 commit that builds against 0.23.7; RTAB-Map's own visual odometry package is not built because the front ends replace it.

Both domains share one launch file, `rtabmap_ros/rtab_sim.launch.py`, and one input shim, `rtabmap_ros/rtab_camera_shim.py`. The shim manufactures what the bags lack: a decoded stereo pair, both `camera_info` messages, and the static transforms from the body frame to the two optical frames, all derived from the VINS configuration directory so the two pipelines read one calibration. A `camera_model` switch selects the pinhole path (simulation) or the fisheye path (real rig); table R.1 lists what the two paths share.

Table R.1: Launch and RTAB-Map settings common to both domains. Values are those set in `rtab_sim.launch.py`; "argument" marks a launch argument with its default.

Setting	Value	Reason
Pose source	TF `odom_frame`→`body_frame`	see table R.2
`approx_sync` / `sync_queue_size`	true / 30	four image topics stamped identically by the shim
`topic_queue_size`	200	raised from 10 for the odometry-topic layout; no longer load-bearing
`wait_for_transform` (argument)	5 s	front-end lag under replay; with a replayed trajectory it is never waited on
`Stereo/DenseStrategy`	1 (SGBM [4])	block matching returned confident garbage on blank floor and walls
`Mem/UseOdomGravity`	true	no IMU reaches RTAB-Map; the front end is gravity-aligned (33 gravity links vs 0)
`RGBD/OptimizeMaxError` (argument)	3.0	upstream default; see table R.2 for its effect with VINS
`Grid/RayTracing`	true	the floor yields no stereo points, so free space exists only by ray tracing
`Grid/NormalsSegmentation`	false	height-based ground split
`Grid/MaxGroundHeight` (`ground_height`)	0.1	measured in `base_frame`
`Grid/MaxObstacleHeight`	2.0 m	ceiling of the obstacle band
`Grid/NoiseFilteringRadius` `Grid/NoiseFilteringMinNeighbors`	0.1, 5	drops isolated speckle
`Grid/RangeMax` (`grid_range_max`)	10.0	simulation default after the range sweep
`Rtabmap/DetectionRate` (`detection_rate`)	1.0 Hz	node creation rate
`loop_closure` (argument)	true	false sets `Rtabmap/LoopThr`=1.0 and `RGBD/ProximityBySpace`=false, keeping visual words so the database still localises
`mode` (argument)	construct	extend / localize switch `Mem/IncrementalMemory`, `Mem/InitWMWithAllNodes`, `RGBD/StartAtOrigin`, `RGBD/CreateOccupancyGrid`
Database wipe	`-d` in construct only	refused in extend / localize; a non-`.db` path is resolved to `<path>/rtabmap.db`
Bag replay	without `--clock`	the bags record `/clock`; a second clock cleared the TF buffer 1724 times

Table R.2: Settings that were measured rather than assumed. Each row is an A/B on one bag with only the named setting changed (engineering record, Sections 6, 6c and 7).

Setting	Condition	Result	Bag
Pose from odometry topic	5-topic sync, 40 Hz odom	7 starvation warnings, 3 nodes, 1 closure	`static_dataset_loop`
Pose from TF	4-topic sync	0 warnings, 28 nodes, 2 closures	same
`Mem/UseOdomGravity` false	default	0 gravity links, map tilted	`dataset_static`
`Mem/UseOdomGravity` true		33 gravity links (one per node), level	same
Extrinsic misread by `cv2.FileStorage`	camera pitched 34°	9 closures; grid precision 42 %, Chamfer p95 1.85 m	`dataset_static_tiny_repeat`
Extrinsic parsed from text, rigidity-checked		72 closures; precision 91 %, p95 0.35 m	same
`RGBD/OptimizeMaxError` 3.0, VINS odometry	drift 0.1–0.2 m	36 closures detected, 0 accepted	same
`RGBD/OptimizeMaxError` 10.0, VINS odometry		36 accepted	same
`RGBD/OptimizeMaxError` 3.0, ground-truth odometry		72 accepted	same

2 Simulation configuration

2.1 Sensor model

The simulated camera is an ideal pinhole, so the shim publishes the raw frames as `image_rect` with no rectification and exits if any distortion term in the calibration is non-zero. Table R.3 gives the values it derives from `vins_fusion_ros2/config/wil_sim/`. Both frames are stamped with the left frame's time, as the ORB-SLAM3 node does, so the two pipelines consume the same pair.

Table R.3: Simulation stereo model as published by the shim (`camera_model:=pinhole`).

Quantity	Value
Resolution	1280×720
$f_x = f_y$; c_x, c_y	640.0; 640.0, 360.0
Distortion	all terms 0
Left / right camera	cam0 / cam1 (cam0 at body $y = +0.0936$; disparity check +11.9 px)
Baseline B; $f_x B$	93.6 mm; 59.9 px m
Body frame	IMU, forward-left-up; body→optical quaternion $(0.5, -0.5, 0.5, -0.5)$
Published rate	4 Hz on the message stamp (3.2 Hz achieved)

2.2 Mapping run as used for the realistic worlds

The simulation mapping runs use the ORB-SLAM3 front end with its own loop closing off, so RTAB-Map is the only component that may close loops. Table R.4 lists the arguments of the command in use against the launch defaults. `ground_height` is 0.29 rather than the default 0.1 because the ORB-SLAM3 node publishes the pose of `base_footprint`, which sits 0.192 m below the IMU body the default was measured for.

Table R.4: Simulation mapping command (realistic worlds, ORB-SLAM3 front end) against the launch defaults.

Argument	Used	Default	Note
`mode`	construct	construct	
`odom_frame` / `body_frame`	`orbslam3_world` / `base_footprint`	`world` / `body`	defaults are the VINS frame names
`ground_height`	0.29	0.1	base_footprint instead of IMU body
`grid_range_max`	8.0	10.0	$f_xB = 59.9$: 1 px error is 1.1 m at 8 m
`detection_rate`	4.0	1.0	keyframe density sets how often a cell is observed
`shim_rate`	4.0	4.0	
`loop_closure`	false	true	closure-free map; closures scored separately
`rviz`	true	false	stock displays, no RTAB-Map plugins built
ORB `loopClosing` (config)	0	–	A/B on the mapping bags: ATE 2.01→0.07 m and 8.82→0.10 m
Bag rate	0.3–0.5	–	keeps the front end's backlog below the TF window

2.3 Front-end coupling

The front end runs in a separate terminal and only its transform is consumed. Running VINS and RTAB-Map together from a bag failed on this laptop (VINS diverged; mechanism not identified), so the VINS localisation levels were measured by replaying a recorded VINS trajectory through `script/gt_tf_publisher.py` instead. The same replay route is the standard procedure on the real rig (section 3.3).

Maps built with this configuration are scored in the Experiment 4 write-up, with their tolerances and sample counts; the scores are not restated here.

3 Real-world configuration

3.1 Sensor model and rectification

The real rig's cameras are 180° Kannala–Brandt [5] fisheyes at 1920×1080, and RTAB-Map's stereo path has no equidistant model, so the shim rectifies every pair onto a virtual pinhole before publishing (`camera_model:=fisheye`). Table R.5 gives the model; figure R.1 shows a rectified pair. The projection matrices are built by hand from `hfov_deg` because OpenCV's own estimate returned $f_x = 0$ for a lens whose corners map to infinity; only its rectification rotations are used.

Table R.5: Real-rig stereo model as published by the shim (`camera_model:=fisheye`, 28 October calibration).

Quantity	Value
Lens model	Kannala–Brandt; $k_1..k_4 = (-3.65\times10^{-2},\ -8.40\times10^{-4},\ 1.77\times10^{-4},\ -3.83\times10^{-4})$
Native intrinsics	$f = 607.5, 609.9$; $c = (958.3, 539.8)$; cam1 reuses cam0's (one mono calibration exists)
Left / right camera	cam1 / cam0 (`left_cam:=1`; cam1 is left since the 19 October remount)
Extrinsic	`camera_parameters_281026`: $B = 89.0$ mm, 0.86°
Virtual pinhole	`hfov_deg` 110: $f_x = 672.2$, $f_xB = 59.8$ px m (the simulation's budget)
`rect_scale` 0.5	960×540, $f_x = 336.1$, $f_xB = 29.9$ px m
Rectification rotations	1.2° (left), 1.7° (right)
Published frames	`cam1_rect_optical`, `cam0_rect_optical`; left bgr8, right mono8

Three invariants are checked at start-up, each of which failed silently at least once while the path was built: the camera named `left_cam` must have the other camera at $+x$ in its optical frame; the two published rectified frames must form a canonical pair, right at $(B, 0, 0)$ with identity rotation, to 10^{-6}; and

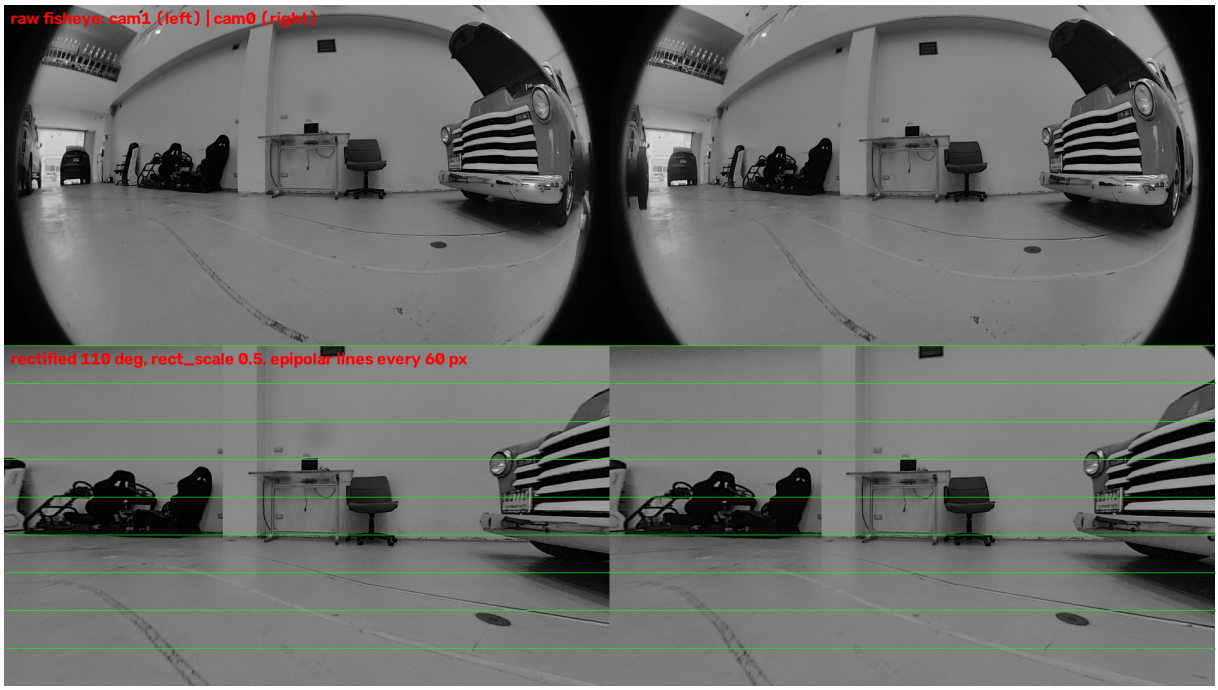


Figure R.1: A realv5 pair before and after rectification (frame 1500, `hfov_deg` 110, `rect_scale` 0.5). The table edge, chair and car grille fall on the same horizontal lines in both halves.

the extrinsic is parsed from the configuration text with comments stripped, because the real-rig file keeps five retired calibrations as commented blocks under the same key and the first match was a pre-remount value.

Table R.6 is the field-of-view trade. A wider virtual view keeps more of the fisheye but lowers the focal length, and the depth error per pixel of disparity is $z^2/(f_x B)$. The default of 110° reproduces the simulation's depth budget.

Table R.6: Virtual field of view against depth budget and epipolar alignment on three realv5 frames at full resolution (cross-checked ORB matches within 2 px of the epipolar line, positive disparity).

HFOV [°]	f_x [px]	$f_x B$ [px m]	on-line matches [%]
90	960	85.5	35
110	672	59.8	53
130	448	39.9	69

3.2 Frames

`imu_link`, the frame the real ORB-SLAM3 launch publishes under `orbslam3_world`, cannot be RTAB-Map's base frame. The IMU extrinsic `IMU.T_b_c1` = diag$(1,-1,-1)$ makes it x-right, y-up, z-backward, and RTAB-Map measures ground height along the base frame's z. A forward-left-up `base_link` is therefore hung under it with a static transform (pitch and yaw 90°, translation $(0,-h,0)$ with h the IMU height above the floor), and the launch takes `base_frame:=base_link` with `ground_height` at its default. h was measured as 0.11 m on 4 November; real-rig runs before that date used a 0.30 m placeholder. Since 4 November the launch publishes this transform itself (`frontend:=orb` or `vins` selects the odometry frames, `publish_base_tf` adds the base frame on the fisheye rig, `imu_height` defaults to the measured value), so no separate transform node is started.

The ORB-SLAM3 node's published world is not gravity-aligned on this rig. It is the active map's frame, which is the camera at map creation, and it differs between runs. On the 28 October run the

body's up axis pointed to world $-y$ for the whole trajectory; on the 31 October run it pointed to $+z$ after a 9.5 s initialisation transient with 91-degree jumps at 2.5, 2.6, 5.3, 5.4 and 9.5 s. The replay publisher therefore measures both from the trajectory file. `world_fix:=auto` rotates the world so the mean body-up axis is $+z$; the axis spreads 1.7° and 2.8° (p95) over the two runs, so the floor is flat enough for this to be a levelling. `start_after_transient:=true` starts 0.5 s after the last jump. The first kept pose is held from the moment the clock runs: a frame with no odometry blocks RTAB-Map for the full `wait_for_transform`, a few such frames backlog the input past the 10 s TF buffer, and three reproductions of that produced databases with zero nodes.

3.3 Two passes, and the compute budget

At the 29 October measurements ORB-SLAM3, the rectifying shim and RTAB-Map did not fit on the laptop together at 1920×1080, so real-rig maps were built in two passes: the front end alone writes `vio.csv`, then the bag is replayed with that trajectory broadcast as TF. On 4 November the two were run together live with the cooling fan on, `rect_scale` 0.5, `detection_rate` 1.0 and `shim_rate` 2.0, and produced a map: the 29 October limit was thermal, not the pipeline. The two-pass route remains the deterministic procedure for A/B comparisons. Table R.7 gives the measured costs that fix the replay settings. At `rect_scale` 0.5 the shim delivers a pair in about 0.4 s and RTAB-Map updates in about 0.3 s; at full resolution the shim alone reached only 1 Hz. The same pipeline on a thermally throttled CPU (92 °C, 0.8–1.3 GHz) was fifteen times slower, which is enough to starve RTAB-Map again, so runs are started on a cool machine.

Table R.7: Measured per-pair costs of the real-rig chain on the laptop (RTX 3070 host), bag at rate 0.5.

Stage	Condition	Time [ms]	Consequence
Shim, decode + remap + publish	1920×1080	1000	≈1 Hz achieved at `shim_rate` 4
Shim	960×540, cool CPU	365–511	keeps up at `shim_rate` 2
RTAB-Map update	960×540, cool CPU	160–320	1 Hz detection rate sustained
RTAB-Map update	same, throttled CPU	4000–4900	input backlog, TF look-ups fail
Shim	same, throttled CPU	4570	

The commands, in start order, are:

```
# 1. the recorded ORB-SLAM3 trajectory as TF
python3 script/gt_tf_publisher.py --ros-args -p use_sim_time:=true \
    -p csv:=output/output_orb/real/<run>/vio.csv \
    -p world_frame:=orbslam3_world -p body_frame:=imu_link \
    -p world_fix:=auto -p start_after_transient:=true
# 2. RTAB-Map with the rectifying shim; frontend:=orb sets the odometry
#    frames and publishes imu_link -> base_link (imu_height 0.11 m default)
ros2 launch ./rtabmap_ros/rtab_sim.launch.py mode:=construct rviz:=true \
    frontend:=orb camera_model:=fisheye left_cam:=1 rect_scale:=0.5 \
    config_dir:=$HOME/wil_project/vins_fusion_ros2/config/<calib> \
    output_path:=map/<name>/ database_path:=map/<name>/<name>.db \
    grid_range_max:=3.0 detection_rate:=1.0 shim_rate:=2.0 \
    loop_closure:=false
# 3. the bag, no --clock
ros2 bag play <bag> --rate 0.5
```

3.4 Calibration validity decides the map

The dense map is only as good as the rectification, and a good ORB-SLAM3 run is no evidence of it. The 31 October recording tracked normally (median map inliers 883 against 735 on 28 October) yet its

map was an unidentifiable radial smear, because the mount had moved: the baseline direction differs by 6.3° from the 28 October calibration. Table R.8 shows the discriminator, the share of feature matches on the epipolar line, for every calibration on disk; figure R.2 shows what the matcher sees. Closures, scene distance and image quality were ruled out first: re-exporting the 31 October database from odometry alone gives the identical cloud (figure R.3), its points lie nearer the path than the 28 October map's, and brightness and sharpness match.

Table R.8: Epipolar alignment of each calibration against the two real recordings. "Stored" rows are the 960×540 pairs RTAB-Map saved; "raw" rows re-rectify full-resolution bag frames (stricter threshold). Compare within a row.

Measurement	realv5 (28 Oct)	realv6 (31 Oct)
stored pairs, 28 Oct calibration	79–87 %	7–22 %
raw, 28 Oct calibration	46 %	1 %
raw, 27 Oct calibration	15 %	15 %
raw, 21 / 26 Oct, 27 Oct_01	0–1 %	0–1 %
raw, self-calibrated from the bag	77 %	43 %
stored pairs, self-calibrated (rebuilt)	–	84 %



Figure R.2: Stored left and right images and the SGBM disparity of one node from the 28 October database (top) and the 31 October database built with the same calibration (bottom). The lower pair is vertically misaligned by about 1.2 px, and the disparity map degrades into streaks.

A self-calibration from the bag restores realv6 to the alignment realv5 has (table R.8, last two rows), and the rebuilt map shows wall lines and object outlines again (figure R.4). `script/stereo_selfcal.py` recovers the rotation and the baseline direction from an essential matrix over fisheye-undistorted matches, with the baseline length copied from a reference file. On realv5 it lands within 0.2° of the chequerboard rotation, which validates the method. The result is kept as `camera_parameters_311026_selfcal/` with a shim configuration in `vins_fusion_ros2/config/wil_311026_selfcal/`; the live `config/wil` is unchanged. A chequerboard calibration of the current mount is still required: the self-calibration leaves a vertical bias of about 1 px and cannot observe the baseline length.

Table R.9 is the measured depth noise of the rectified pair, which fixes the range settings: a plane fitted to the floor points of several nodes, median distance of the points from that plane per range bin. At `rect_scale` 0.5 the depth is grid-cell accurate to about 1.5 m and within the 25 cm scoring tolerance to about 3–4 m. `grid_range_max` is therefore 3 m for real-rig maps, against 10 m in simulation; a range limit for viewing the cloud is set in the viewer or the export tool and does not touch the database.

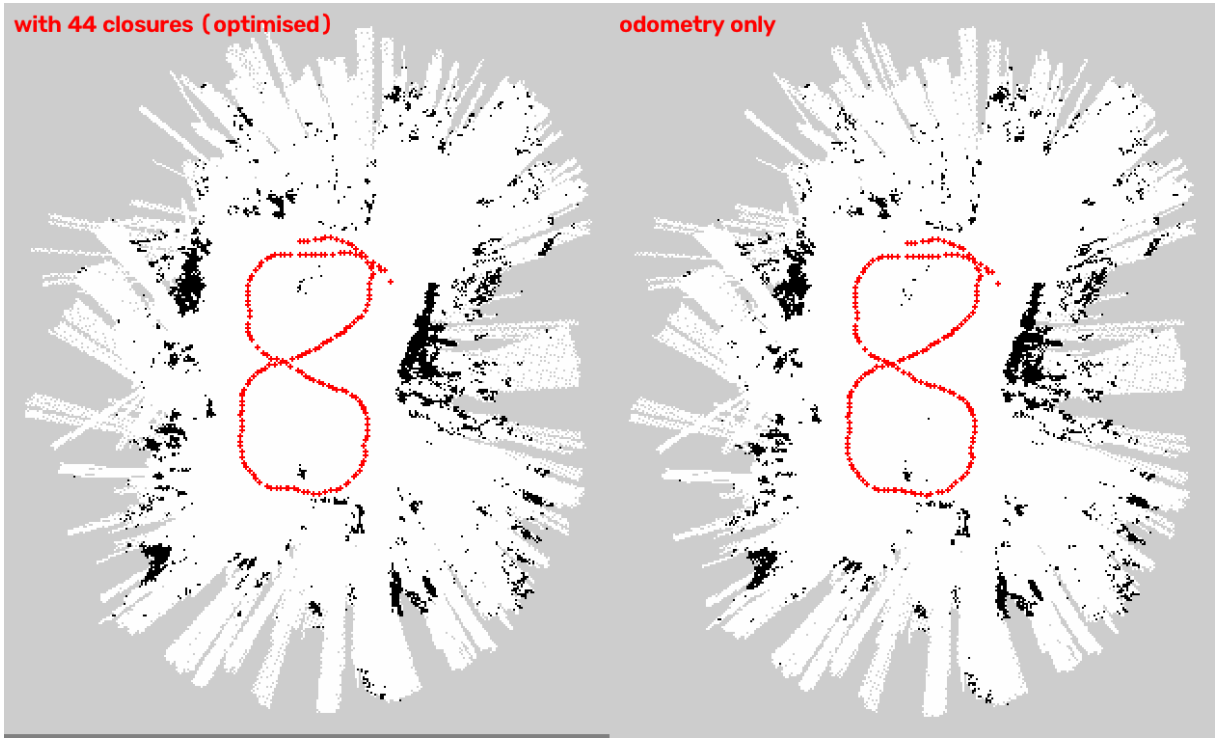


Figure R.3: The 31 October grid exported with its 44 accepted closures (left) and from odometry alone (right); node poses in red. The two are the same map, so the closures are not what smeared it.

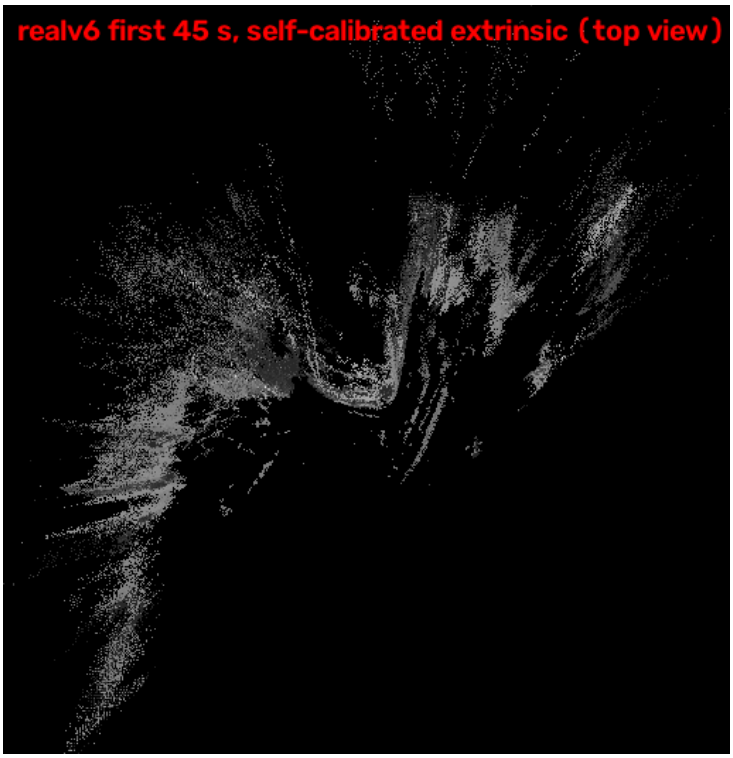


Figure R.4: Top view of the dense cloud from the first 45 s of the 31 October recording rebuilt with the self-calibrated extrinsic (36 nodes). Structure returns; residual ray streaks remain.

Table R.9: Floor-plane residual of SGBM depth per range bin on the stored 960×540 pairs ($f_x B = 29.9$), median over 5–8 nodes per database.

Range [m]	realv5, 28 Oct [cm]	realv6, self-cal. [cm]	realv6, misrectified [cm]	1 px theory [cm]
0.5 – 1	0.9	4.1	3.3	2
1 – 1.5	2.6	6.3	5.6	5
1.5 – 2	6.5	4.9	12.6	10
2 – 3	6.6	7.0	29.5	21
3 – 4	13.8	13.4	32.6	41
4 – 6	60.8	24.6	72.2	84

3.5 Real recordings and databases

Table R.10 registers the two real recordings that reached RTAB-Map and the databases built from them. Neither recording has an independent ground truth, so these databases carry node and link counts and the calibration check, not a map-quality score. Figure R.5 is the 28 October grid.

Table R.10: Real recordings and RTAB-Map databases at the cutoff. Loop closures are RTAB-Map's accepted global and local-space links.

Recording	Content	ORB run replayed	Databases
`dataset_realv5_000`, 28 Oct 15:31, 134.6 s, 4038 pairs at 30 Hz, mount of the 28 Oct calibration	triangular route past parked cars; `/clock` recorded; no `camera_info`, TF or ground truth	`realv5_imu_cbgroup_reliable_r05_newIMU_TO_CAM_TF` (1366 rows, no reset after the transient)	`map/real/real.db`: 87 nodes, 2 trivial start closures, stored pairs 79–87 % on-line; cars identifiable
`dataset_realv6_000`, 31 Oct 19:00, 220 s, 6601 pairs, mount moved after 28 Oct	figure-of-eight in open hall	`dataset_realv6_000/loopclosing_rate1` (3083 rows, ORB loop closing on, 0 closures fired)	`map/real_311026_rate05/`: 168 nodes, 1 closure; `map/real_311026_rate05_lc/`: 189 nodes, 34 + 10 closures; both 7–22 % on-line (unusable); self-calibrated rebuild of the first 45 s: 36 nodes, 84 %

3.6 Failure signatures

A database with zero nodes means the input chain is dead, and the console says where. Table R.11 lists the signatures met while bringing the real-rig path up, in the order to check them. A database with nodes but no grid points instead at a grid or base-frame setting.

3.7 Open items

- Chequerboard calibration of the current mount, then a check of the next recording with `script/stereo_selfcal.py` before any map is built. Calibrate first, record second.
- The real ORB-SLAM3 configuration carries `loopClosing:1`; set it to 0 for a mapping run, as in simulation, and back to 1 to load an Atlas.
- No real-site ground truth: map quality on the real rig can only be assessed by the calibration check, the node and link counts and visual inspection until an independent reference exists.
- The real-rig mapping bags are the ones ORB-SLAM3 was evaluated on; localize and extend modes have not been run on a real database.

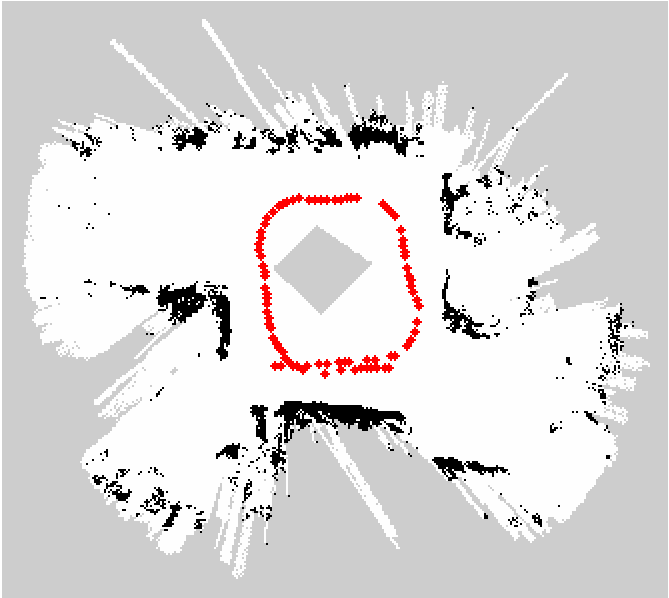


Figure R.5: Occupancy grid of `map/real/real.db` (28 October, 87 nodes, 5 cm cells, `grid_range_max` 6) with node poses in red. Walls and parked cars appear as occupied cells; the radial streaks are ray-traced free space from far stereo points, which the 3 m range limit of table R.9 removes.

Table R.11: Zero-node failures of the real-rig chain and their signatures.

Cause	Console signature	Fix in place
Shim exits at start-up: `hfov_deg:=110` parsed as an integer against a double parameter	Python traceback in the launch output within 2 s; RTAB-Map idle	launch hands floats; shim declares the parameter dynamically typed
Frames before the first odometry transform	"Did not receive data since 5 seconds" alternating with one "extrapolation into the past" every 10 s	replay holds the first kept pose from clock start (`hold_before_start`)
Orphaned shim from an earlier run publishing the same topics	"Messages … arrived out of order"; per-pair time ten times normal	`ps -eopid,lstart,args|grep shim` before every run; start test processes in their own session
Thermal throttling	RTAB-Map update 4–5 s; then as above	start cool; bag rate 0.5

4 Reporting rules

- Launch and shim: `vslam_navigation/launch/rtabmap.launch.py` and `vslam_navigation/scripts/rtab_camera_shim.py` (moved from the `rtabmap_ros` fork root on 4 November; a forwarding stub keeps the old command); replay `script/gt_tf_publisher.py`; calibration check `script/stereo_selfcal.py`. Superproject `23c6c0b`; the `rtabmap_ros` fork at `d18dc87` with the 26 October to 2 November changes to these files uncommitted at the cutoff.
- Simulation measurements (table R.2, table R.4): `rtabmap_ros/MEMORY_CLAUDE/README.md` Sections 5, 6, 6c, 7 and 7c; the ORB loop-closing A/B in `output/compare/simulation/lc-mapping-ab/`.
- Real-rig measurements (tables R.6 to R.9): the same record, Section 7d; figures and their data under `output/compare/real/rtab-config/`.
- Databases: `map/real/real.db`, `map/real_311026_rate05/`, `map/real_311026_rate05_lc/`; the self-calibrated rebuild was a 45 s verification segment and is not kept as a map.
- Real recordings:
  `dataset_realv5_000` (then on the flash drive; since 3 November 2026 all bags live under `/media/ambushee/NewVolume/wil_rescue/`, where realv5 is absent) and `dataset/dataset_realv6_000`;
  ORB runs under `output/output_orb/real/`.

References

[1] Labbé, M. and Michaud, F., 2019, "RTAB-Map as an Open-Source Lidar and Visual Simultaneous Localization and Mapping Library for Large-Scale and Long-Term Online Operation," **Journal of Field Robotics**, Vol. 36, No. 2, pp. 416–446.

[2] Campos, C., Elvira, R., Gómez Rodríguez, J.J., Montiel, J.M.M. and Tardós, J.D., 2021, "ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial, and Multi-Map SLAM," **IEEE Transactions on Robotics**, Vol. 37, No. 6, pp. 1874–1890.

[3] Qin, T., Li, P. and Shen, S., 2018, "VINS-Mono: A Robust and Versatile Monocular Visual-Inertial State Estimator," **IEEE Transactions on Robotics**, Vol. 34, No. 4, pp. 1004–1020.

[4] Hirschmüller, H., 2008, "Stereo Processing by Semiglobal Matching and Mutual Information," **IEEE Transactions on Pattern Analysis and Machine Intelligence**, Vol. 30, No. 2, pp. 328–341.

[5] Kannala, J. and Brandt, S.S., 2006, "A Generic Camera Model and Calibration Method for Conventional, Wide-Angle, and Fish-Eye Lenses," **IEEE Transactions on Pattern Analysis and Machine Intelligence**, Vol. 28, No. 8, pp. 1335–1340.